Lecture plan · for the lecturer

Bias, variance and cross-validation

Data science · University · Lecture 2

Time
60 min

Materials
Projector or board; The lecture slides and the seminar sheet from the kit; Phones or laptops with Darsi, for the lab

Learning goals

- Decompose expected error.
- Estimate error with cross-validation.
- Prevent leakage in student work.

Activity: Find the leak

1. Warm-up (5 min): Show two models' training and test errors and ask students to vote, then reveal the real-world results.
2. Teach (15 min): Derive the decomposition and fit polynomials of rising degree live.
3. Hunt (20 min): Pairs compare CV scores across the leaky and clean notebooks and find each leak.
4. Fix (15 min): Rewrite one notebook with a Pipeline and grouped folds, and compare the honest score.
5. Wrap-up (5 min): Students write the muddiest point of the lecture and one question; read them before next week.

Discussion

- Why does training error always fall?
- What is the unit of independence in your data?
- How will your brief stop leakage?

Exit ticket

- Name one leak and the line of code that fixes it.

Try it in your next lecture
Hand out two short notebooks, one with leakage and one without, and ask pairs to explain why their cross-validation scores differ before you say anything.

Seminar sheet · for students

Bias, variance and cross-validation

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Name _____ Group _____ Date _____

1 **A student standardises all the data, then runs 5-fold cross-validation. Is the estimate honest?**

- ◯ No, the test folds leaked into the scaling
- ◯ Yes, if k is at least 5
- ◯ Yes, scaling doesn't use the labels

2 **Work out the cross-validation estimate.**

Fold errors 10, 12, 8, 14, 11 give a 5-fold CV error of _____ . The final test set is used _____ times for choosing the model.

Word bank: 55 · 11 · 0 · 5

3 **Which of these leak information into a cross-validation estimate? Choose all that apply.**

- ☐ Selecting the 20 most correlated features on all the data, then running CV
- ☐ Several rows per student, split randomly across folds
- ☐ Fitting the scaler inside each training fold
- ☐ Filling missing values with the mean of all rows first

4 **Adding more training data mainly reduces a model's variance, not its bias.**

- ◯ True
- ◯ False

5 **Match each symptom to its most likely cause.**

Training and test error both high	• • • •	High variance
Training error low, test error high	• • • •	Too few data or folds
CV score far better than the later real-world score	• • • •	High bias
CV error changes a lot between random splits	• • • •	Leakage

Answers · for the lecturer

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1 **No, the test folds leaked into the scaling**

Any statistic computed on the test fold leaks into the model.

2 **Fold errors 10, 12, 8, 14, 11 give a 5-fold CV error of 11. The final test set is used 0 times for choosing the model.**

(10 + 12 + 8 + 14 + 11) / 5 = 11; the final test set is never used for choices, or its score becomes optimistic.

3 **Selecting the 20 most correlated features on all the data, then running CV; Several rows per student, split randomly across folds; Filling missing values with the mean of all rows first**

Feature selection and imputation on all rows, and splitting one student's rows across folds, all let test information in; scaling inside each fold is correct.

4 **True**

More data stabilises a flexible fit; a model that is too simple stays wrong however much data it sees.

5 **Training and test error both high — High bias**
Training error low, test error high — High variance
CV score far better than the later real-world score — Leakage
CV error changes a lot between random splits — Too few data or folds

Asking students to diagnose from symptoms builds the judgement a formula test can't reach.

For you to think about

Think of a student project you marked last year. Could leakage have inflated its score? How would your project brief prevent it now?